

Pratheek Shetty

Technical Lead — Distributed Systems | Backend Platforms | Real-Time Data Infrastructure

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PROFESSIONAL SUMMARY

Technical Lead with 6+ years of experience in System Design, engineering large-scale distributed systems, real-time data platforms, and cloud-native infrastructure. Expert in microservices architecture, event-driven stream processing, IoT communication systems, and high-availability backend platforms leveraging Kafka, Apache Flink, MQTT, WebSockets, Apache Pinot, PostgreSQL, and AWS/GCP ecosystems. Currently leading cross-functional engineering teams at Simple Energy, architecting data pipelines that process billions of telemetry events while strategically optimizing infrastructure to reduce cloud expenditure by 44%. Proven track record establishing core software patterns, leading technical execution across matrixed teams, and scaling production vehicle architectures.

CORE EXPERTISE

System Design & Architecture

Distributed Systems Infrastructure

Real-Time Ingestion & WebSockets

Data Lakehouse & Iceberg Tables

Relational Databases & PostgreSQL

Event-Driven Stream Processing

Infrastructure Cost Optimization

Cross-Functional Engineering Leadership

Connected Vehicle & IoT Platforms

Generative AI & LLM Orchestration

CRM Integration & CDC Pipelines

TECHNICAL SKILLS

Data & Streaming	Kafka, Apache Flink, Apache Spark, EMQX MQTT Broker, MQTT Protocol, WebSockets, Apache Pinot, ClickHouse, Redis, PostgreSQL, SQL, Apache Iceberg Tables, Delta Lake, Amazon Redshift, AWS DMS, AWS Glue, AWS Athena
Backend & Languages	Node.js, TypeScript, JavaScript, Express.js, GraphQL, REST APIs, Electron.js, LangGraph, LLM Orchestration, AI Support Automation, System Design, SOLID Principles
Frontend Platform	React.js, Next.js, Redux, D3.js, Micro-Frontends, Responsive UI Architecture, Material UI
Cloud & Infrastructure	AWS (Lambda, EC2, ECS, Cognito, IoT Core, S3, SNS, SQS, RDS, AppSync), GCP (Kubernetes/ GKE, Compute Engine, GCR, GCS), CI/CD, Git, Infrastructure as Code (IaC)

KEY INFRASTRUCTURE ACHIEVEMENTS

- Cost Optimization:** Reduced cloud infrastructure expenditure by 44% through data telemetry optimization, storage tiering via Apache Iceberg, resource consolidation, and architectural improvements.
- Scale Engineering:** Architected connected vehicle telemetry pipelines processing billions of telemetry data points, maintaining 500B+ analytical data records in Apache Pinot for sub-second fleet analytics.
- Notification Throughput:** Built event-driven notification infrastructure capable of handling millions of critical alert notifications per day at sub-second latencies.
- Team Leadership:** Led and coordinated a 23-member cross-functional engineering team spanning backend, frontend, mobile apps, DevOps, data engineering, and data science functions.

PROFESSIONAL EXPERIENCE

Technical Lead — Simple Energy

Bengaluru, India | Mar 2022 – Present

Connected vehicle platform powering real-time telemetry, OTA updates, and analytics infrastructure for thousands of production electric vehicles.

- Platform Strategy & System Design:** Owned the complete architectural lifecycle and high-level system design of the connected vehicle platform, supporting large-scale telemetry ingestion and real-time analytics pipelines.
- Stream Processing & Ingestion:** Designed a dual-pipeline ingestion model utilizing Kafka, EMQX MQTT Broker, and WebSockets to execute sub-second telemetry analytics across streaming vehicle datasets.

- **Data Lakehouse Engineering:** Constructed scalable data lakehouse structures using AWS Glue catalogs, Apache Iceberg tables, and AWS Athena to query transactional petabyte-scale telemetry lakes with zero server management overhead.
- **Serverless & Compute Optimization:** Built real-time processing worker blocks using event-driven AWS Lambda functions to ingest, filter, and normalize vehicle metrics dynamically before long-term persistent layout storage.
- **Database Management:** Engineered transactional data stores and relational schemas using PostgreSQL, ensuring optimized query performance, database normalization, and fault-tolerant storage tiering for high-volume writes.
- **OTA Platforms:** Engineered secure Over-the-Air (OTA) firmware delivery backend infrastructures enabling cryptographic validation and automated remote fleet updates.
- **AI Platforms & Systems:** Architected a production enterprise generative AI system utilizing Amazon Redshift, AWS DMS, and S3 to orchestrate organizational knowledge indexing, delivering context-aware natural language insights across corporate data lakes.
- **AI Support Orchestration:** Built custom telemetry intelligence and support automation agents using LangGraph and LLM orchestration frameworks for automated vehicle troubleshooting and analytics workflows.
- **Operational Tooling:** Created low-latency diagnostic tools with Electron.js, optimizing factory-floor automated testing, provisioning validation, and hardware configuration routines.
- **Integration Architecture:** Designed and implemented a low-latency business integration connector layer facilitating high-fidelity LMS and Salesforce integrations; orchestrated real-time CDC (Change Data Capture) pipelines to stream updates directly from transactional RDS layers to external CRM endpoints.
- **Engineering Standards:** Established design patterns across Node.js microservices, conducted regular architecture reviews, and instituted automated CI/CD safeguards to maximize service reliability.
- **Frontend Platform Engineering:** Led the frontend architecture strategy for mission-critical internal analytics, vehicle diagnostics dashboards, and management portals using React.js, Next.js, and Redux.

Software Engineer — Tech Active

Bengaluru, India | Oct 2021 – Feb 2022

- **Product Architecture:** Directed system design and implementation of backend APIs for an audio licensing and contract collaboration ecosystem leveraging Node.js and scalable database schemas.
- **Dispatch Integration:** Designed internal service management and real-time logistics tracking engines using Next.js and WebSockets, linking live driver status assignments with automated billing workflows.
- **Performance Tuning:** Optimized database indexing in PostgreSQL and server-side caching mechanics on multi-tenant REST APIs, stabilizing backend response times under heavy concurrent traffic loads.

Frontend Developer — Filmyscoop

Bengaluru, India | Apr 2020 – Sep 2021

- **IoT Integration:** Engineered an IoT-enabled smart lighting controller platform backend interfaces and web control applications from concept to full production deployment within a 2-month window.
- **UI Standardization:** Constructed modular, reusable interface components in React.js, focusing heavily on client-side state optimization and asset performance.

TECHNICAL WRITING & THOUGHT LEADERSHIP

Published technical deep-dives on distributed architectures, real-time analytics pipelines, and engine mechanics on Medium:

- **Building a Scalable Connected Vehicle Platform** with HMI, EMQX, Kafka, Flink, and Pinot.
- **Evolving Our Connected Vehicle Data Platform:** Smart Dual-Pipeline Architecture.
- **Apache Pinot Optimization** for Real-Time IoT Telemetry Analytics.
- **Building a Production-Grade Support AI Agent** using modern LangGraph structures.
- **Handling Millions of Notifications at Scale** using event-driven architectures.
- **The Tale of Node.js:** Deconstructing Call Stack execution mechanics and the Event Loop.

EDUCATION

Mangalore Institute of Technology & Engineering (MITE)

2016 – 2020

Bachelor of Technology in Computer Science and Engineering